

# Raphael Karamagi

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## EDUCATION

### Duke University

Durham, NC

*B.S.E. Electrical & Computer Engineering, B.S. Computer Science, GPA: 3.83*

*Aug. 2025 – May 2029*

**Selected Coursework:** Computer Architecture, Signals & Systems, Data Structures & Algorithms, Probability, Linear Algebra, Multivariable Calculus

## TECHNICAL SKILLS

**Languages:** Python, Java, C/C++, R, SQL, JavaScript

**ML & Data:** PyTorch, TensorFlow, scikit-learn, OpenCV, pandas, NumPy

**Frameworks & Tools:** Flask, React, Node.js, PostgreSQL, Git, Docker, Linux/Bash, OAuth

## EXPERIENCE

### Software Developer

Aug 2026 – Present

*Duke OIT – Innovation Co-Lab*

*Durham, NC*

- Selected to join Duke OIT's developer team for the 2026–27 academic year, supporting internal tool & infrastructure development and technical workshops for the Duke community

### Software Engineer Intern

May 2026 – July 2026

*Duke OIT – Code+ Program*

*Durham, NC*

- Built an ML model security & safety scanning pipeline—spanning model weights, dependencies, and inference red-teaming—producing risk-scored reports before 50+ ML models are deployed to Duke faculty and students.
- Developed efficacy tooling through public benchmarks, and Duke specific 'LLM-as-judge suites.'
- Deployed as a containerized platform backed by Postgres, GitLab CI pipelines, and user authentication.

### Software Engineer Intern

July 2024 – Aug. 2024

*IntelliSOFT Consulting*

*Nairobi, Kenya*

- Implemented REST API endpoints with role-based access in OpenMRS EMR; led deployment to 20+ hospitals across South Sudan and 3 Nairobi pilot sites with 98% uptime.

## PROJECTS

### Stock Prediction Platform v2 | *Python, PyTorch, JavaScript, C++, FinBERT*

Jan. 2026 – Present

- Co-led 5-person team building a dual-model (LSTM + NLP) prediction system across 15 tickers.
- Architecting v2 around an event-driven backtester with order book simulation, transaction cost modeling, and walk-forward validation

### Compost Now Wash Automation | *Arduino, C/C++, MOSFET, Embedded*

Jan. 2026 – Apr. 2026

- Led team designing automated shutoff for industrial bin washers, reducing waste water and energy for 150+ bins washed per day.
- Designed embedded control circuit (Arduino Nano + 12V solenoid via MOSFET switches, LED beacons, user inputs) and wrote non-blocking C++ firmware.

### ASL Computer Vision Classifier | *PyTorch, OpenCV*

Oct. 2025 – Dec. 2025

- Built a dual-model CV system classifying 29 ASL gestures from 87K+ images (71% test accuracy); designed Landmark NN (~10K params) for real-time, local CPU inference with 100% hand-detection accuracy.

## LEADERSHIP & ACTIVITIES

### Duke Applied Machine Learning – Senior Engineer

Sep. 2025 – Present

- Mentor junior members through the AITP program; lead Stock Prediction Platform team.

### National Society of Black Engineers (NSBE) – Active Member, Duke Chapter

Aug. 2025 – Present

- Represented Duke at Fall 2025 Regional Conference and 2026 Annual Convention (Baltimore, MD).

## HONORS & AWARDS

5th Globally, Cambridge AS-Level Economics | UKMT Senior Math Challenge Silver (2024) | SAT 1540/1600